

MIMIC-IV Database

INPUT: EHR data



Note



radiology



discharge



Hospitalization



patient



preception



labevent



pharmacy



microbio



procedure



diagnosis



admission

STEP 1: Event & Patient Filtering

⚠ abnormal flags

✅ complete notes

🕒 > 15 visits



355 Patients 6,999 Visits 314,294 Events

STEP 2: Event Stream Construction

$P^{***}.json$

$\mathcal{V}_1$

Visit 001

$\mathcal{V}_2$

Visit 002

$n > 15$

$\mathcal{V}_n$

Visit n

E_i^{adm} 09/12/07 11:44

admission CC, HR...

E_i^1 09/12/07 12:35

labs CBC, BMP...

...

E_i^k t_i^k

$a_i^k \in \mathcal{A}$ p_i^k, r_i^k

...

E_i^{dis} 09/13/07 08:20

discharge notes



Note Summary $N_i = N_i^{adm} + N_i^{dis}$

STEP 3: Memory Dataset Generation

History visit: $\mathcal{V}_i$ Current visit: $\mathcal{V}_j$ Timestamp: T

$P^{***}-\mathcal{V}_i-E_1$ 09/12/07 19:35

IMAGING Targeted liver ultrasound to target right hepatic lobe hyperechoic...

$P^{***}-\mathcal{V}_i-E_2$ $P^{***}-\mathcal{V}_i-E_3$... $P^{***}-\mathcal{V}_i-E_m$

$P^{***}-\mathcal{V}_{i+1}-E_1$ $P^{***}-\mathcal{V}_{i+1}-E_2$... $P^{***}-\mathcal{V}_2-E_n$

...

$P^{***}-\mathcal{V}_{j-1}-E_1$ $P^{***}-\mathcal{V}_{j-1}-E_2$... $P^{***}-\mathcal{V}_{j-1}-E_p$

...

Assistant 09/12/07 15:27 order_imaging, "modality": "xray", "target": "chest"...

User 09/12/07 17:45 imaging_results: ... The lungs volumes are low...

Assistant 09/12/07 15:27 order_imaging, "modality": "ultrasound", "target": ...

...

$E_i^k \rightarrow \{(\text{assistant}, \{a_i^k, t_i^k, p_i^k\}), (\text{user}, \{f_k, t_k'\})\}$ for all k with $t_k < T$



Event Memory



Contextual Memory



Note Memory

OUTPUT: Memory datasets